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FLOOR TOOL ATTACHMENT FOR USE WITH VACUUM CLEANER

Abstract

An attachment for a vacuum cleaner includes a head, a brush roll, an electric motor, and an attachment conduit. The head includes a suction opening, a brush roll cavity, and a battery cavity defined therein. The brush roll cavity is in fluid communication with the suction opening. The battery cavity has an elongate shape and a first longitudinal axis. The brush roll is rotatably coupled to the head. A majority of the brush roll is disposed within the brush roll cavity. The brush roll is rotatable about a rotational axis. The electric motor is disposed within the head. The electric motor drives the brush roll. The attachment conduit is in fluid communication with the suction opening. The attachment conduit is coupled to the head and has a second longitudinal axis. Each of the first longitudinal axis and the second longitudinal axis is angled relative to the rotational axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/370,093, filed Jul. 8, 2021, which claims priority to U.S. Provisional Patent Application No. 63/049,947, filed Jul. 9, 2020, the entire contents of each of which is incorporated by reference herein.

FIELD

[0002] The present disclosure relates to vacuum cleaner accessories and attachments. More particularly, the present disclosure relates to a battery-powered floor tool attachment for use with a vacuum cleaner.

SUMMARY

[0003] In one aspect, the present disclosure relates to an attachment for use with a vacuum cleaner. The attachment includes a head, a brush roll, an electric motor, and an attachment conduit. The head includes a suction opening defined therein, a brush roll cavity defined therein, and a battery cavity defined therein. The brush roll cavity is in fluid communication with the suction opening. The battery cavity has an elongate shape and a first longitudinal axis. The brush roll is rotatably coupled to the head. A majority of the brush roll is disposed within the brush roll cavity. The brush roll is rotatable about a rotational axis. The electric motor is disposed within the head. The electric motor drives the brush roll. The attachment conduit is in fluid communication with the suction opening. The attachment conduit is coupled to the head and has a second longitudinal axis. Each of the first longitudinal axis and the second longitudinal axis is angled relative to the rotational axis.

[0004] In another aspect, the present disclosure relates to an attachment for use with a vacuum cleaner. The attachment includes a head, a brush roll, an electric motor, and an attachment conduit. The head includes a battery cavity defined therein. The brush roll is disposed in the head. The brush roll is rotatable about a rotational axis. The electric motor is disposed in the head. The electric motor drives the brush roll. The attachment conduit is coupled to the head. The attachment conduit is arranged between the battery cavity and the electric motor in a direction parallel to the rotational axis.

[0005] In another aspect, the present disclosure relates to an attachment for use with a vacuum cleaner. The attachment includes a head, a power tool battery pack, a brush roll, a first electric motor, and an attachment conduit. The power tool battery pack is removably coupled to the head. The brush roll is rotatably coupled to the head. The first electric motor is disposed within the head. The first electric motor drives the brush roll. The attachment conduit is coupled to the head. The power tool battery pack is usable with the attachment to power the first electric motor when coupled to the head. The power tool battery pack is also usable with a separate power tool to power a second electric motor of the separate power tool when coupled to the separate power tool.

[0006] Features and aspects of the disclosure will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a perspective view of a floor tool attachment for use with a vacuum cleaner, according to one embodiment disclosed herein.

[0008] FIG. **2** is a front elevation view of the floor tool attachment of FIG. **1**.

[0009] FIG. **3** is a rear elevation view of the floor tool attachment of FIG. **1**.

[0010] FIG. **4** is a top plan view of the floor tool attachment of FIG. **1**.

[0011] FIG. **5** is a bottom plan view of the floor tool attachment of FIG. **1**.

[0012] FIG. **6** is a left side elevation view of the floor tool attachment of FIG. **1**.

[0013] FIG. **7** is a right side elevation view of the floor tool attachment of FIG. **1**.

[0014] FIG. **8** is a perspective view of a battery for use with the floor tool attachment of FIG. **1** as well as other example power tools.

DETAILED DESCRIPTION

[0015] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

[0016] With reference to FIG. **1**, an embodiment of a floor tool attachment **100** for use with a vacuum cleaner is shown. The floor tool attachment **100** includes a head **102**, a brush roll **104**, an electric motor **106**, and an attachment conduit **108**. The head **102** has two opposite ends with the brush roll **104** extending at least partially therebetween.

[0017] The head **102** may be a nozzle having a suction opening **110** defined therein. The suction opening **110** is placed adjacent a surface to be cleaned during operation of the floor tool attachment **100** and draws up dust, dirt, or other debris via suction provided by an attached vacuum cleaner (not shown). In the illustrated embodiment, the suction opening **110** is an elongate slot and extends along a majority of the length of the head **102**.

[0018] The brush roll **104** is coupled to the head **102** and is configured to rotate at least partially within the head **102** for engaging a surface to be cleaned to facilitate debris collection. In some embodiments, the brush roll **104** extends a majority of distance between the two opposite ends of the head **102**. In the illustrated embodiment, a brush roll cavity **112** is also defined in the head **102**, and the brush roll cavity **112** is in fluid communication with the suction opening **110**. The brush roll cavity **112** is located downstream from the suction opening **110** in the direction of airflow. The brush roll **104** is rotatably coupled to the head **102** in the brush roll cavity **112** and rotates about a rotational axis RA1. In some embodiments, at least a majority of the brush roll **104** is disposed in the brush roll cavity **112**. The brush roll **104** may extend through the suction opening **110** such that engagement of the brush roll **104** with the surface to be cleaned cooperates with suction airflow into suction opening **110** to facilitate debris collection. In the illustrated embodiment, the brush roll **104** is disposed such that only a portion of surface-engaging members of the brush roll **104** extend beyond the suction opening **110** to engage the surface to be cleaned, for instance, to agitate and loosen ingrained dirt in a carpet. In some embodiments, the brush roll **104** may be a more conventional brush roll design having a dowel with one or more rows of helically arranged bristles that extend radially outwardly from the dowel. In such embodiments, the bristles would be considered the surface-engaging members.

[0019] The head **102** also includes a motor cavity **114** defined therein. The electric motor **106** is disposed in the motor cavity **114** and is operably connected to the brush roll **104** to facilitate rotation thereof. In some embodiments, the electric motor **106** is disposed nearer a second end of the head **102** than a first end of the head **102**. In the illustrated embodiment, the electric motor **106** is positioned such that the driveshaft **116** of the electric motor **106** extends generally parallel to the

rotational axis RA1 of the brush roll 104 and from the electric motor 106 toward the second end of the head 102. The electric motor 106 is indirectly coupled to the brush roll 104 via a drive belt 118 coupled to each of the driveshaft 116 and the brush roll 104. In this manner, the electric motor 106 drives the brush roll 104. In some embodiments, a drive chain, drive gears, or some other transmission arrangement may be used instead of the drive belt 118.

[0020] The head 102 further includes a battery cavity 120 defined therein. In some embodiments, the battery cavity 120 is disposed nearer the first end of the head 102 than the second end of the head 102. The battery cavity 120 extends along a first longitudinal axis LA1. In some embodiments, at least a portion of the battery cavity 120 is recessed relative to at least one outer surface of the head 102. In other embodiments, at least a portion of the battery cavity 120 is defined outwardly from the head 102. In the illustrated embodiment, the floor tool attachment 100 further includes a battery mount 122 extending outwardly from the head 102 along the first longitudinal axis LA1. The battery mount 122 can be a variety of shapes and/or sizes to fit an appropriate battery pack 124. In the illustrated embodiment, the battery mount 122 is configured to receive a battery pack 124 that is also compatible with a variety of other battery-powered tools such as, for instance, power drills, power saws, blowers, work lights, or the like. The battery mount 122 is a generally hollow projection having a mount sidewall 126. In cross-section, the mount sidewall 126 is a generally circular sidewall surrounding the battery cavity 120. In some embodiments, therefore, the battery cavity 120 is a cylindrical battery cavity. In other embodiments, the battery cavity 120 may be a rectangular prism, a triangular prism, and elliptical prism, or the like. In some embodiments, the first longitudinal axis LA1 is generally normal to the rotational axis RA1. In some embodiments, the normal angle is also a perpendicular angle. That is, the first longitudinal axis LA1 and the rotational axis RA1 intersect instead of simply being oriented in planes that are perpendicular to each other. In other embodiments, however, the first longitudinal axis LA1 is oblique or parallel relative to the rotational axis RA1. In the illustrated embodiment as shown in FIG. 7, the battery mount 122 extends along the first longitudinal axis LA1 a first distance D1 away from the rotational axis RA1. In some embodiments, the battery cavity 120 also extends the first distance D1 away from the rotational axis RA1 due to the battery cavity 120 being defined by the battery mount 122. The battery pack 124 is slidably received in the battery cavity 120 along the first longitudinal axis LA1.

[0021] In some embodiments, the floor tool attachment 100 further includes a switch 128 disposed on or in the head 102. The switch 128 selectively electrically communicates the battery pack 124 with the electric motor 106. In the illustrated embodiment, the switch 128 is a push button, but other embodiments may include different types of switches such as, for instance, a tab, a knob, or the like. The switch 128, in some embodiments, may directly control the electrical communication between the battery pack 124 and the electric motor 106. In some embodiments, the switch 128 is disposed between the battery cavity 120 and the first end of the head 102. In some embodiments, the switch 128 is actuatable in a direction parallel to the first longitudinal axis LA1 and a second longitudinal axis LA2 (described in more detail below).

[0022] The floor tool attachment 100 may further include a printed circuit board (PCB) 130. In the illustrated embodiment, the PCB 130 can function as a controller to control the transfer of electrical energy from the battery pack 124 to the electric motor 106, the speed of the electric motor 106, the illumination of one or more lights (not shown), or the like. Further, some embodiments may include one or more sensors (not shown) that may send signals to the PCB 130 corresponding to brush roll speed, brushroll current, air pressure within the floor tool attachment 100, carpet height, movement speed of the floor tool attachment 100, or the like. In some embodiments, the PCB 130 has a length that extends parallel to the rotational axis RA1 of the brush roll 104.

[0023] The floor tool attachment 100 also includes an attachment conduit 108 for transporting dirty air containing debris drawn into the head 102 to a collection chamber (not shown) of the vacuum cleaner. The attachment conduit 108 extends outwardly from the head 102 along the second

longitudinal axis LA2. In some embodiments, the attachment conduit **108** is arranged between the battery cavity **120** and the electric motor **106** in a direction parallel to the rotational axis RA1. The attachment conduit **108** includes a conduit sidewall **132** and a conduit opening **134** defined by the conduit sidewall **132**. The conduit sidewall **132** can be a variety of shapes and/or sizes to fit a corresponding wand, hose, or nozzle of a corresponding vacuum cleaner. In the illustrated embodiment, the conduit sidewall **132** is generally circular in cross-section. The attachment conduit **108** is in fluid communication with the suction opening **110** and the brush roll cavity **112**. The floor tool attachment **100** is configured such that dirt is picked up through the suction opening **110**, passes through the brush roll cavity **112**, passes through the attachment conduit **108**, and exits the conduit opening **134**. Because the attachment conduit **108** is sized and/or shaped to couple to a component of the vacuum cleaner (not shown), the dirt then is captured by, for instance, the dirt cup of the vacuum cleaner after exiting the conduit opening **134** and traveling through the nozzle, wand, and/or hose of the vacuum cleaner. In some embodiments, the second longitudinal axis LA2 is generally normal to the rotational axis RA1 of the brush roll **104**. In some embodiments, the normal angle is also a perpendicular angle. That is, the second longitudinal axis LA2 and the rotational axis RA1 intersect instead of simply being oriented in planes that are perpendicular to each other. In the illustrated embodiment, the second longitudinal axis LA2 is also generally parallel with the first longitudinal axis LA1 such that the battery cavity **120** and the attachment conduit **108** extend generally parallel to each other away from the rotational axis RA1. As shown in FIG. 7, the attachment conduit **108** extends away from the rotational axis RA1 a second distance D2 along the second longitudinal axis LA2. In some embodiments, the second distance D2 is longer than the first distance D1. For instance, the first distance D1 may have a length that is more than one third of the length of the second distance D2. In some embodiments, the first distance D1 has a length that is more than one half of the length of the second distance D2. In some embodiments, the attachment conduit **108** is wider than the battery cavity **120**.

[0024] With reference to FIG. 8, the battery pack **124** is shown as being connectable to a variety of tools including the floor tool attachment **100**. Other tools usable with the battery pack **124** include, for instance, a handheld vacuum cleaner **136**, a drill/driver **138**, a soldering iron **140**, a palm impactor **142**, a saw **144**, or the like. In this sense, the battery pack **124** is a typical power tool battery pack usable with a variety of power tools compatible with the same battery pack **124**. The battery pack **124** powers an electric motor, heating element, speaker, light, and/or other battery-powered components of the various power tools when coupled thereto.

[0025] Various features of the disclosure are set forth in the following claims.

Claims

1. An attachment for use with a vacuum cleaner, the attachment comprising: a head including: a suction opening defined in the head as an elongate slot defining a plane, a brush roll cavity defined in the head, the brush roll cavity in fluid communication with the suction opening, and a battery cavity defined in the head and configured to support a battery pack therein, the battery cavity having an elongate shape and a first longitudinal axis; a brush roll rotatably coupled to the head and configured to engage a ground surface to be cleaned through the suction opening, a majority of the brush roll disposed within the brush roll cavity, the brush roll rotatable about a rotational axis; an electric motor disposed within a motor cavity of the head; the electric motor configured to drive the brush roll; a controller located within the head and configured to control a transfer of electrical energy from the battery pack to the electric motor; and an attachment conduit in fluid communication with the suction opening, the attachment conduit coupled to the head and having a central axis extending along the attachment conduit in an airflow direction through the attachment conduit; wherein a length of the attachment is taken along the rotational axis of the brush roll, a height of the attachment is taken along the central axis of the attachment conduit, and a depth of the

attachment is perpendicular to the length and the depth, wherein the length is greater than the height and the height is greater than the depth.

2. The attachment of claim 1, wherein the elongate shape of the battery cavity is defined at least partially by a battery mount extending outwardly from the head.

3. The attachment of claim 2, wherein the battery cavity provides an electrical connection between the controller and the battery pack positioned within the battery cavity.

4. The attachment of claim 3, wherein the motor cavity and the brush roll cavity collectively define an interior volume of the head and the battery cavity extends inwardly such that at least a portion of the battery cavity is positioned lower than a portion of the interior volume when the suction opening is positioned adjacent the ground surface, and the battery cavity extends outwardly with the battery mount such that at least a portion of the battery cavity is positioned above the interior volume when the suction opening is positioned adjacent the ground surface.

5. The attachment of claim 4, wherein the battery mount includes a generally circular sidewall.

6. The attachment of claim 4, wherein the battery mount extends away from the rotational axis a first distance along the first longitudinal axis, the attachment conduit extends from the rotational axis a second distance along the central axis, and the first distance is more than one third of the second distance.

7. The attachment of claim 6, wherein the first distance is more than one half of the second distance.

8. The attachment of claim 1, wherein the first longitudinal axis and the central axis are generally normal to the rotational axis.

9. The attachment of claim 1, wherein the battery cavity slidably receives the battery pack along the first longitudinal axis.

10. The attachment of claim 1, further comprising a switch disposed on the head, the switch configured to selectively electrically couple the electric motor with the battery pack.

11. The attachment of claim 1, wherein the attachment conduit is integrally formed as a single component with the head.

12. An attachment for use with a vacuum cleaner, the attachment comprising: a head including: a suction opening defined in the head as an elongate slot defining a plane, a brush roll cavity defined in the head, the brush roll cavity in fluid communication with the suction opening, and a battery cavity defined in the head and configured to support a battery pack therein, the battery cavity having an elongate shape and a first longitudinal axis; a brush roll rotatably coupled to the head and configured to engage a ground surface to be cleaned through the suction opening, a majority of the brush roll disposed within the brush roll cavity, the brush roll rotatable about a rotational axis; an electric motor disposed within a motor cavity of the head; the electric motor configured to drive the brush roll; a controller located within the head and configured to control a transfer of electrical energy from the battery pack to the electric motor; an attachment conduit in fluid communication with the suction opening, the attachment conduit coupled to the head and having a central axis extending along the attachment conduit in an airflow direction through the attachment conduit; wherein each of the first longitudinal axis and the central axis is angled transverse to the plane of the suction opening, wherein the elongate shape of the battery cavity is defined at least partially by a battery mount extending outwardly from the head, wherein the battery cavity provides an electrical connection between the controller and the battery pack positioned within the battery cavity, and wherein the motor cavity and the brush roll cavity collectively define an interior volume of the head and the battery cavity extends inwardly such that at least a portion of the battery cavity is positioned lower than a portion of the interior volume when the suction opening is positioned adjacent the ground surface, and the battery cavity extends outwardly with the battery mount such that at least a portion of the battery cavity is positioned above the interior volume when the suction opening is positioned adjacent the ground surface.

13. The attachment of claim 12, wherein the first longitudinal axis and the central axis are generally

normal to the rotational axis.

14. The attachment of claim 13, wherein the electric motor includes a driveshaft extending generally parallel to the rotational axis.

15. The attachment of claim 12, wherein the battery cavity slidably receives the battery pack along the first longitudinal axis.

16. The attachment of claim 12, further comprising a switch disposed on the head, the switch configured to selectively electrically couple the electric motor with the battery pack.

17. The attachment of claim 12, wherein the battery mount includes a generally circular sidewall.

18. The attachment of claim 12, wherein the battery mount extends away from the rotational axis a first distance along the first longitudinal axis, the attachment conduit extends from the rotational axis a second distance along the central axis, and the first distance is more than one third of the second distance.

19. The attachment of claim 18, wherein the first distance is more than one half of the second distance.

20. An attachment for use with a vacuum cleaner, the attachment comprising: a head including: a suction opening defined in the head as an elongate slot defining a plane, a brush roll cavity defined in the head, the brush roll cavity in fluid communication with the suction opening, and a battery cavity defined in the head and configured to support a battery pack therein, the battery cavity having an elongate shape defined at least partially by a battery mount extending outwardly from a remainder of the head; a brush roll rotatably coupled to the head and configured to engage a ground surface to be cleaned through the suction opening, a majority of the brush roll disposed within the brush roll cavity, the brush roll rotatable about a rotational axis; an electric motor disposed within a motor cavity of the head; the electric motor configured to drive the brush roll; a controller located within the head and configured to control a transfer of electrical energy from the battery pack to the electric motor; and an attachment conduit integrally formed as a single component with the head and in fluid communication with the suction opening, the attachment conduit having a central axis extending along the attachment conduit in an airflow direction through the attachment conduit, wherein the motor cavity and the brush roll cavity collectively define an interior volume of the head and the battery cavity extends inwardly such that at least a portion of the battery cavity is positioned lower than a portion of the interior volume when the suction opening is positioned adjacent the ground surface, and the battery cavity extends outwardly with the battery mount such that at least a portion of the battery cavity is positioned above the interior volume when the suction opening is positioned adjacent the ground surface.
